



Abstract

THIS PROJECT PRESENTS THE DESIGN AND IMPLEMENTATION OF A LOW-COST, EFFICIENT, AND RELIABLE ANALOG GAS DETECTION SYSTEM FOR SAFETY MONITORING. THE SYSTEM USES AN MQ-2 OR MQ-135 GAS SENSOR ALONG WITH AN LM393 COMPARATOR TO DETECT HAZARDOUS GASES SUCH AS LPG, METHANE, AND CARBON MONOXIDE. VISUAL AND AUDIBLE ALERTS ARE PROVIDED VIA LEDS AND A BUZZER. THE SYSTEM IS DESIGNED TO OPERATE BOTH WITH AND WITHOUT AN ARDUINO, MAKING IT VERSATILE FOR REAL-WORLD APPLICATIONS IN HOMES AND INDUSTRIES.



Introduction

GAS LEAKAGE POSES SERIOUS RISKS OF FIRE, EXPLOSION, AND HEALTH HAZARDS. TRADITIONAL DETECTION SYSTEMS ARE OFTEN EXPENSIVE AND COMPLEX. THIS PROJECT AIMS TO DEVELOP AN AFFORDABLE AND SIMPLE ANALOG GAS DETECTION SYSTEM THAT CAN PROMPTLY ALERT USERS IN CASE OF GAS LEAKAGE, THEREBY ENHANCING SAFETY IN RESIDENTIAL AND INDUSTRIAL ENVIRONMENTS.

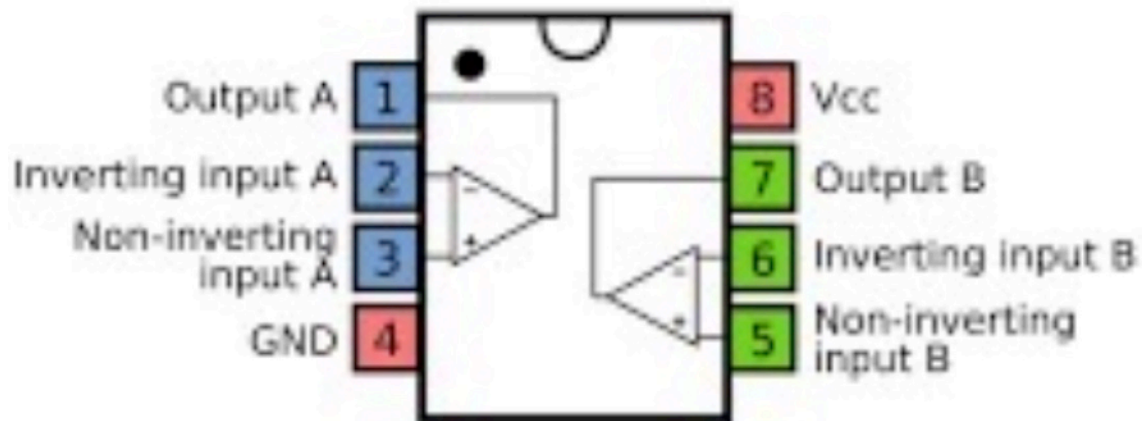


Components Used

1. LM393 COMPARATOR IC
2. MQ-2 / MQ-135 GAS SENSOR
3. RESISTORS (220Ω, ETC.)
4. LEDS (GREEN, YELLOW, RED)
5. BUZZER
6. BUCK CONVERTER (5V REGULATOR)
7. ARDUINO UNO (FOR TESTING AND DEMONSTRATION)
8. BREADBOARD & JUMPER WIRES

Working Principle

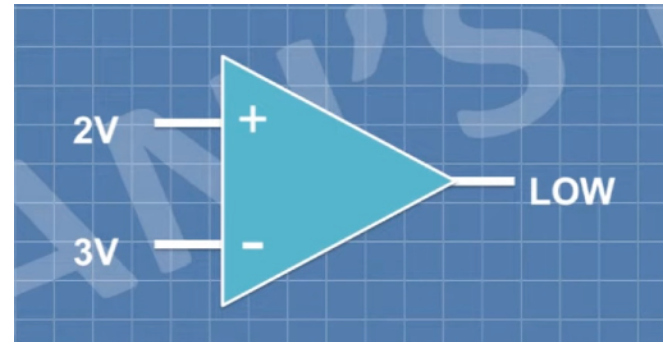
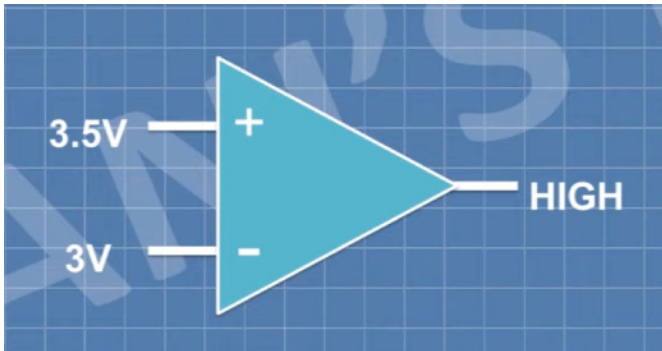
- THE GAS SENSOR DETECTS THE PRESENCE OF HAZARDOUS GASES AND PRODUCES AN ANALOG VOLTAGE OUTPUT.
- THIS VOLTAGE IS COMPARED WITH A REFERENCE VOLTAGE USING THE LM393 COMPARATOR.
- IF THE SENSOR VOLTAGE EXCEEDS THE REFERENCE, THE COMPARATOR OUTPUT GOES HIGH, ACTIVATING THE RED LED AND BUZZER.
- GREEN AND YELLOW LEDS INDICATE SAFE AND WARNING STATES RESPECTIVELY.



LM393 DIP package

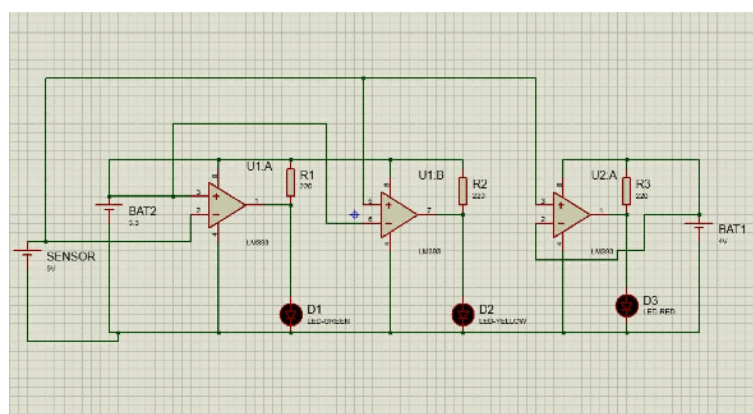
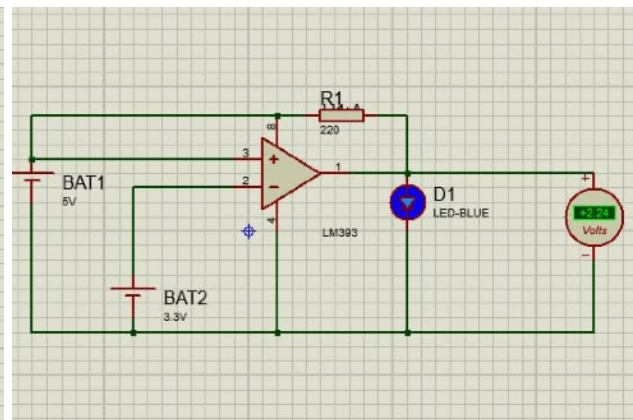
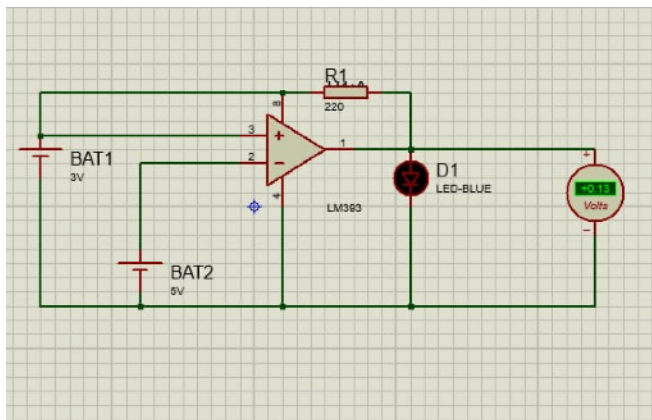
Pin description:

1. **Output 1** – Output of comparator 1 (open-collector, needs pull-up)
2. **Inverting input 1 (-)** – Comparator 1 input
3. **Non-inverting input 1 (+)** – Comparator 1 input
4. **GND** – Connect to 0V
5. **Non-inverting input 2 (+)** – Comparator 2 input
6. **Inverting input 2 (-)** – Comparator 2 input
7. **Output 2** – Output of comparator 2 (open-collector)
8. **VCC** – Connect to supply voltage (e.g., 5V)

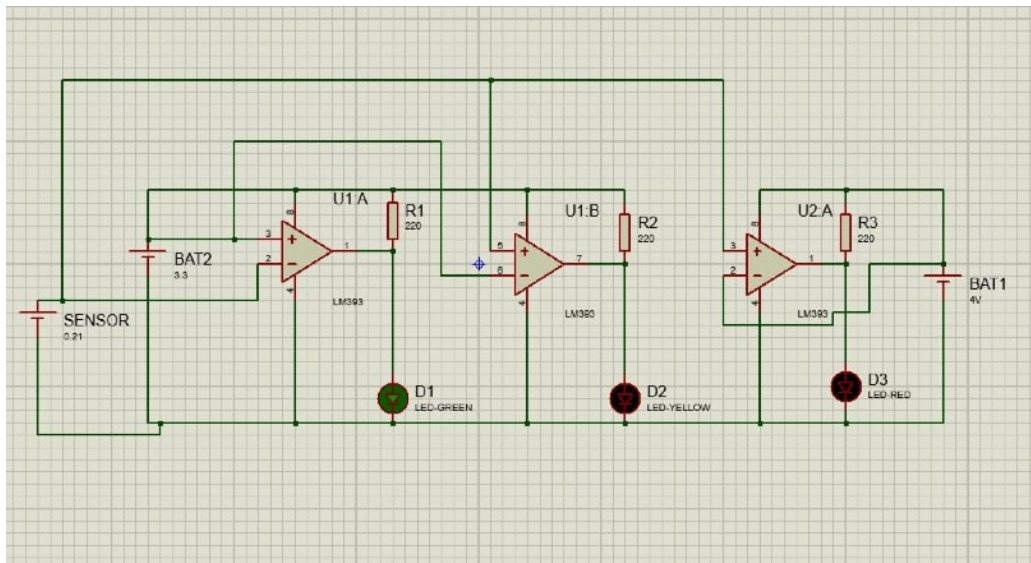


- **LM393 output is open-collector**, so it cannot go high on its own.
- To get a **high output**, connect a **pull-up resistor** (e.g., 4.7kΩ–10kΩ) from the output pin to VCC.
- **How it works:**
 - If the non-inverting input (+) > inverting input (-), the output transistor **turns off**, and the pull-up resistor pulls the output **high**.
 - If (+) < (-), the transistor **conducts**, pulling the output **low**.

Basically: **pull-up resistor makes LM393 output high when the transistor is off.**



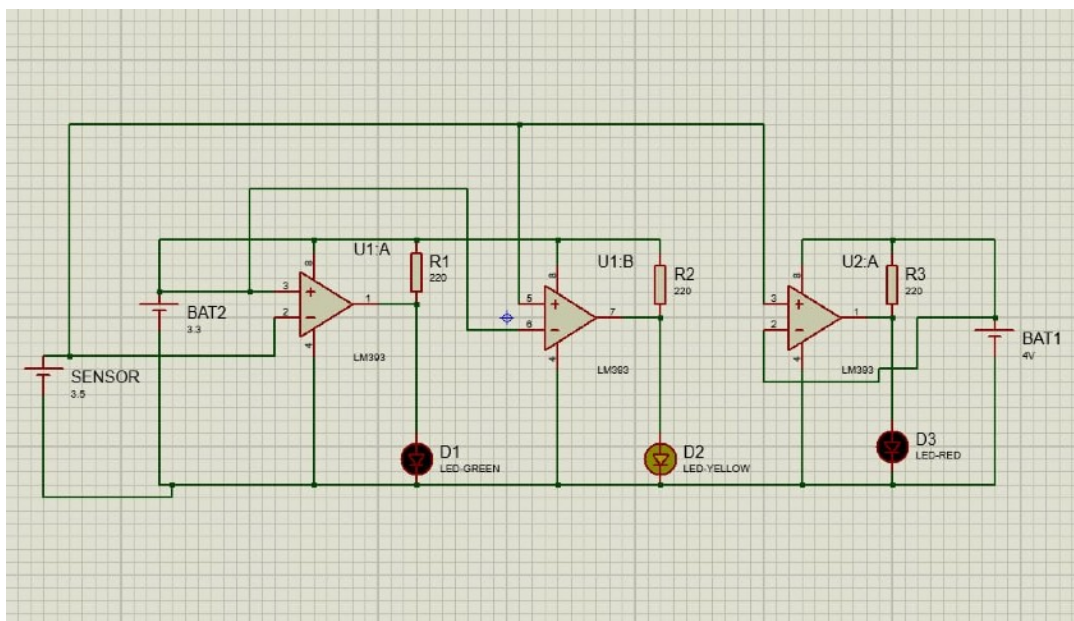
3 LM393 Comparators control 3 LEDs based on gas sensor voltage:



1st Comparator (Green LED):

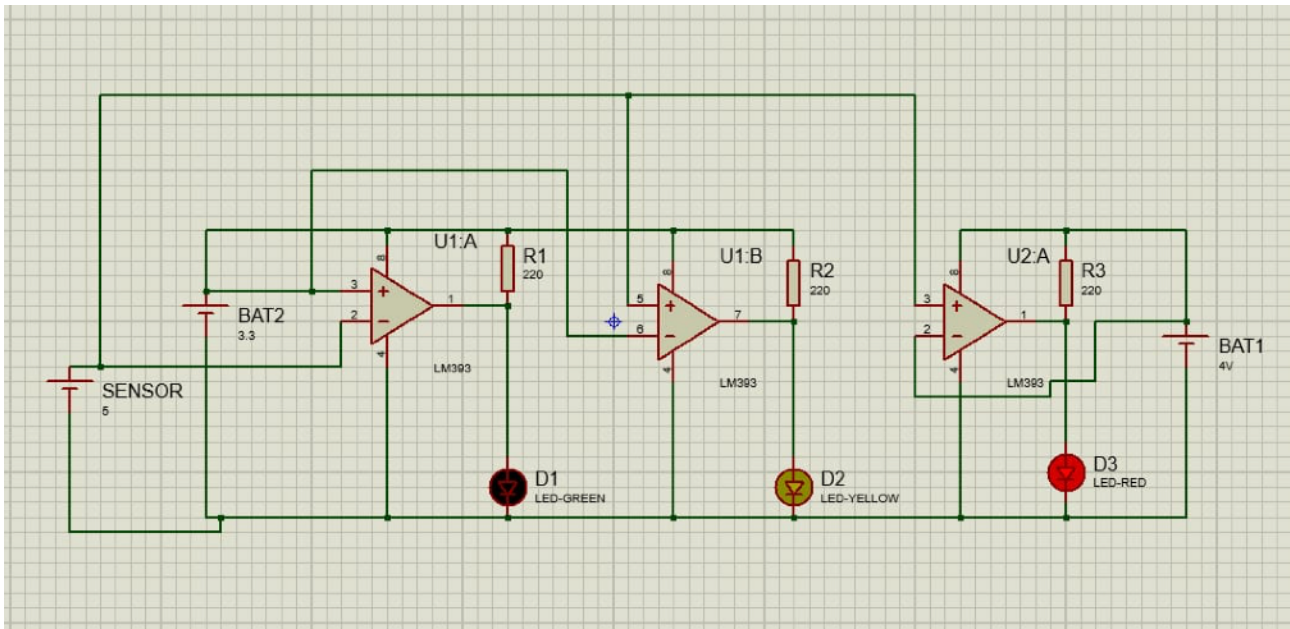
REFERENCE: 3.3V(NON INVERTING)

- SENSOR VOLTAGE (INVERTING) > 3.3V → OUTPUT LOW → GREEN LED OFF
- MEANING: GAS DETECTED ABOVE SAFETY LEVEL



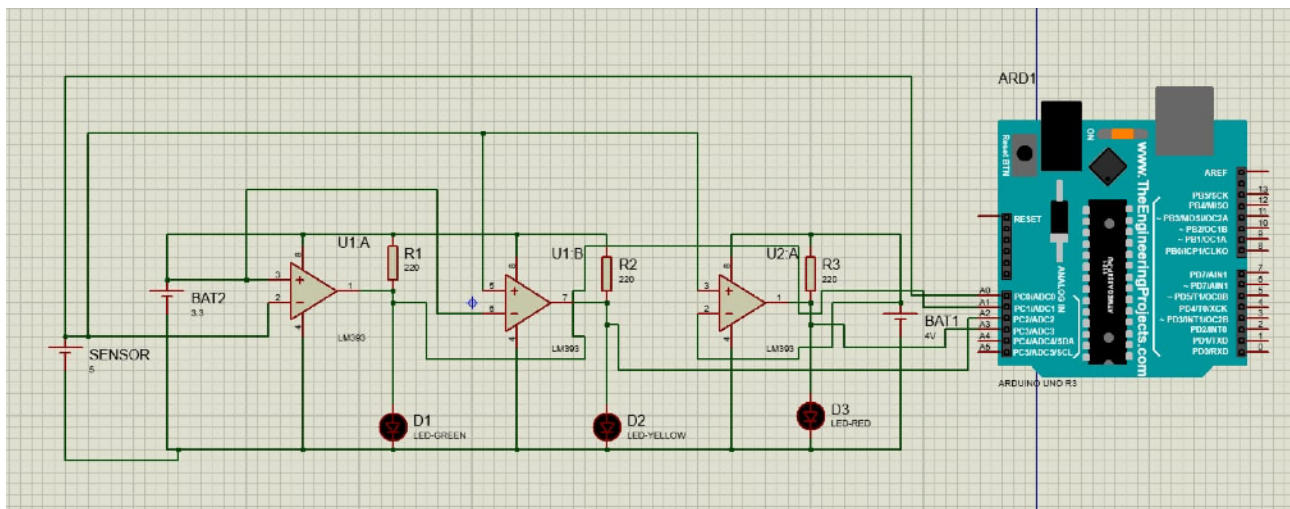
2nd Comparator (Yellow LED):

- REFERENCE: 3.3V (INVERTING)
- SENSOR VOLTAGE (NON INVERTING) > 3.3V BUT < 5V → OUTPUT HIGH → YELLOW LED ON
- MEANING: MODERATE GAS LEVEL WARNING



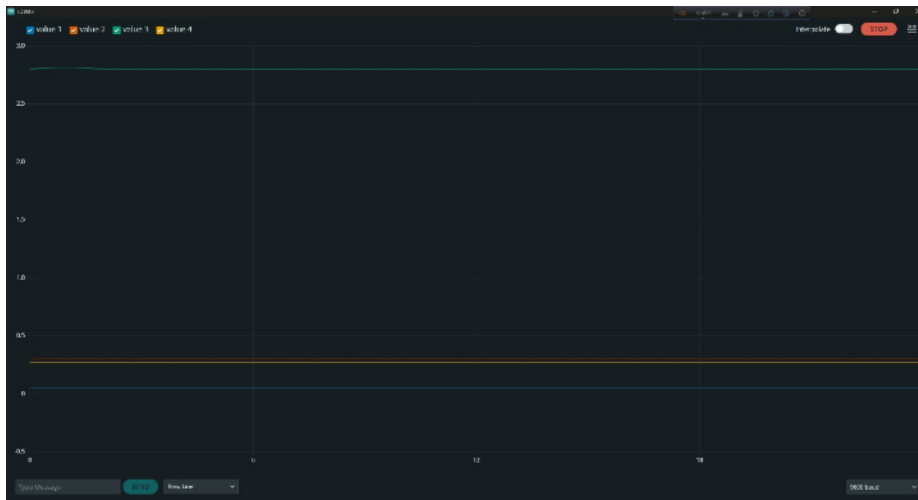
3rd Comparator (Red LED + Buzzer):

- Reference: 4V (Inverting)
- Sensor voltage(Non Inverting) > 4V → Output HIGH → Red LED ON + Buzzer
- Meaning: High gas level danger



Arduino UNO

Used to demonstrate how the comparator and circuit work. Helps to easily monitor sensor output and fault conditions. Makes testing and debugging simple. The circuit can also work independently without Arduino.

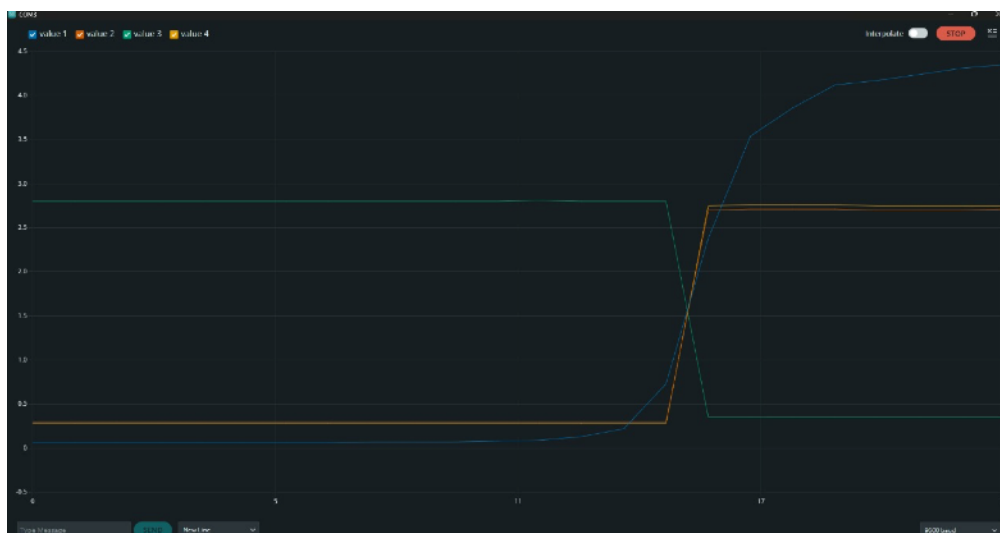


Experimental Data Collection:

- ARDUINO SERIAL MONITOR USED FOR REAL-TIME DATA VISUALIZATION
- SERIAL PLOTTER GRAPHICALLY DISPLAYS GAS CONCENTRATION TRENDS
- ANALOG VALUES (0-1023) AND VOLTAGE (0-5V) MONITORED SIMULTANEOUSLY
- DATA LOGGED AT 1000MS INTERVALS FOR ACCURATE RESPONSE ANALYSIS

Benefits of Serial Monitoring:

- ✓ Real-time visualization of sensor response
- ✓ Calibration verification of threshold values
- ✓ Performance analysis of response/recovery times
- ✓ Troubleshooting sensor or circuit issues
- ✓ Data documentation for experimental validation



Problem Faced:

1 Previous Circuit Issues:

Tried using 555 timer + OP-AMP + BJT + Diode circuit.

Switching was not reliable.

Diode voltage drop too low → incorrect triggering.

2 Sensor Problems:

One sensor got completely damaged.

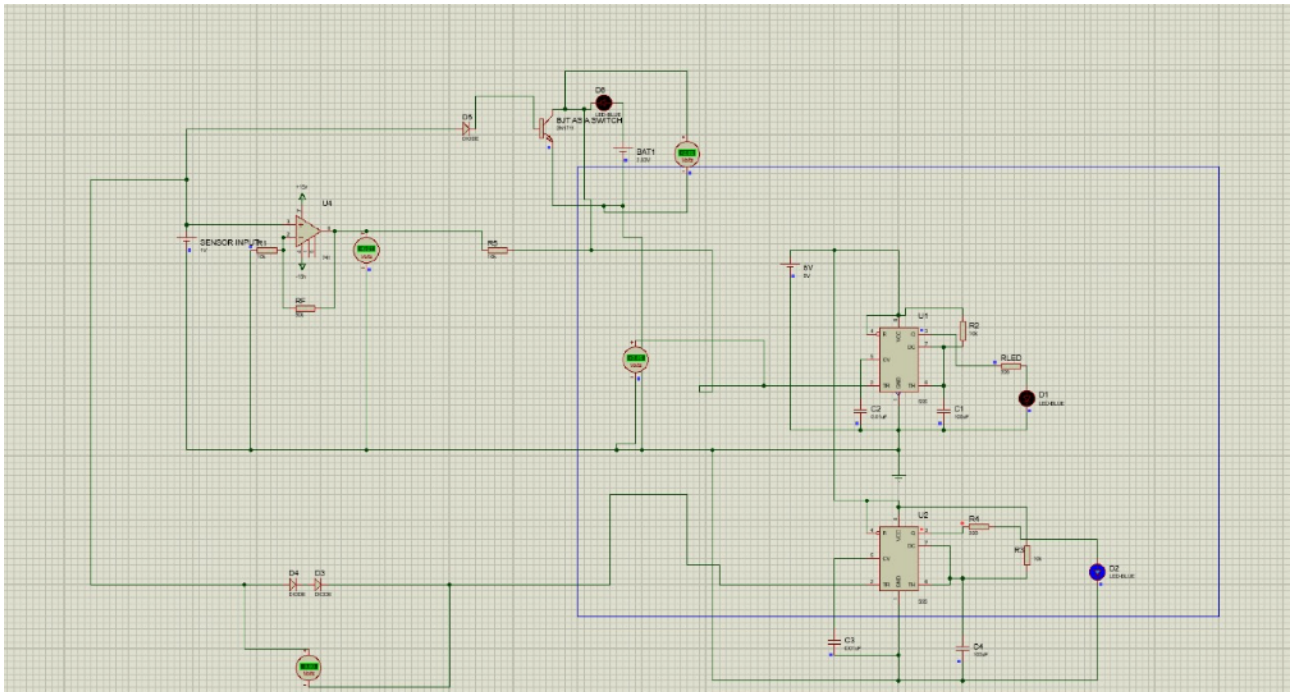
Another sensor had high initial voltage that kept increasing on each test.

Sensor replaced with a better model for accurate detection.

3 New Circuit Design:

LM393 comparator-based circuit used for stable HIGH/LOW output.

LED & buzzer trigger reliably.



Applications:

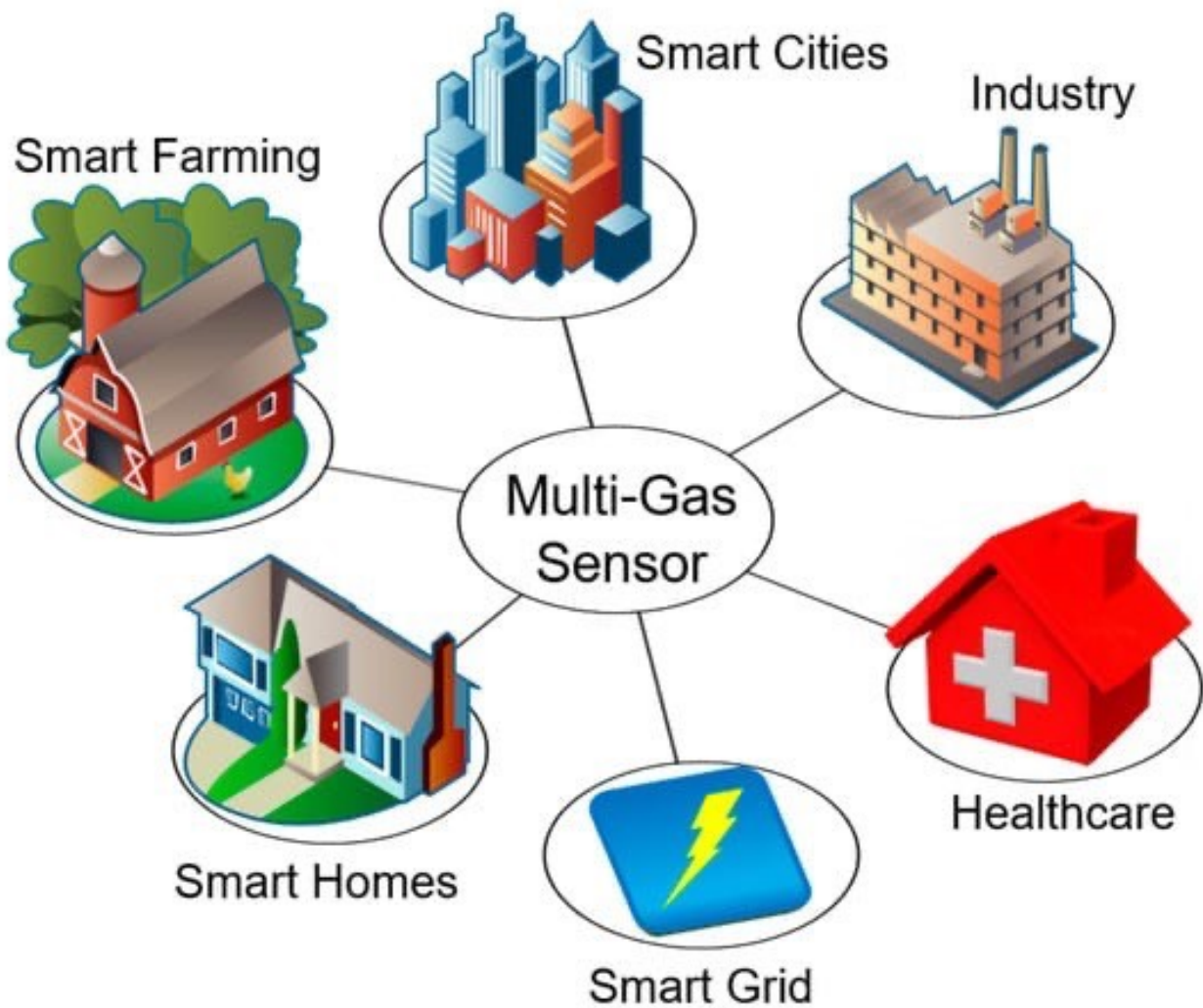
“This system can be used in homes, industries, and laboratories for safety purposes.”

LPG leakage detection

Industrial gas monitoring

Fire prevention systems

Smart home safety devices



💡 Advantages & Limitations

Advantages:

- **Low Cost:** The circuit uses inexpensive and easily available components.
- **Simple Design:** Easy to construct and understand, making it suitable for beginners.
- **Real-Time Response:** Provides immediate output based on changing input conditions.

Limitations:

- **Requires Calibration:** The sensor or comparator circuit needs periodic calibration for accurate results.
- **Environmental Sensitivity:** Performance may vary due to humidity, temperature, or dust conditions.

🧩 Conclusion

The LM393-based circuit is a simple, low-cost, and efficient solution for real-time signal comparison or sensor-based applications. Although it requires calibration and is slightly affected by environmental factors, its fast response and easy implementation make it an excellent choice for small-scale and educational projects.

Acknowledgement

I would like to express my sincere gratitude to my course teacher **Md. Sharif Uddin** and **Proctor Sir Md. Khairul Alam** for their constant guidance, valuable suggestions, and continuous support throughout this project.

I am also thankful to my **group members** and the **Project Lab, University of Asia Pacific (UAP)**, for their cooperation and assistance during the development and testing phases of my project.

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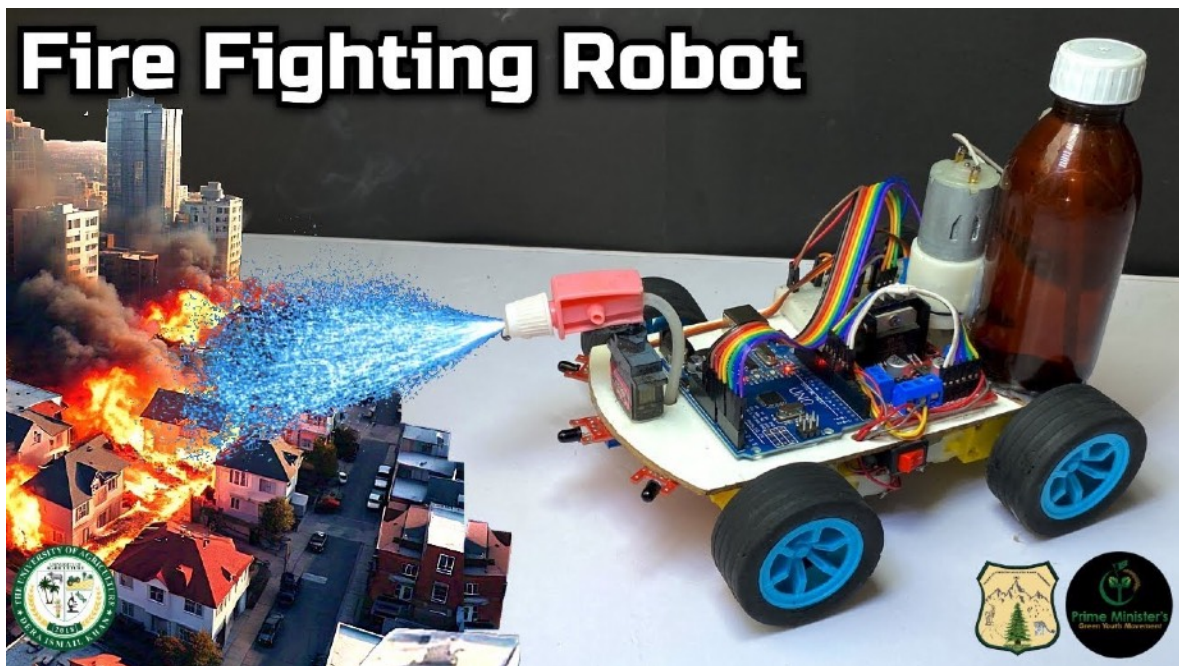
Contribution

Component selection, oscilloscope, Data Collection testing, Documentation, result analysis,

Circuit design & simulation(Protius),
PCB design(EasyEDA), Verify Circuit Working(Aurdno)&
Coding(Aurdno IDE),Presentation preparation(Keynote),
Assembly circuit.

Presentation preparation, report writing, block
diagram, Assembly Circuit.

Documentation, simulation, result analysis, Coding
Result Analysis



Future Work (Fire Fighting BOT)

- Develop a small moving vehicle capable of detecting **gas leaks** and **fire sources**.
- The vehicle will **send alerts** when it detects any danger.
- Use **special frequencies or sound waves** to help **reduce or extinguish fire**.
- Connect the system to a **mobile phone or computer** for real-time alert monitoring.
- Enhance the design for **home and industrial safety applications**.
- Improve **sensor accuracy** and **response time** for better performance.



